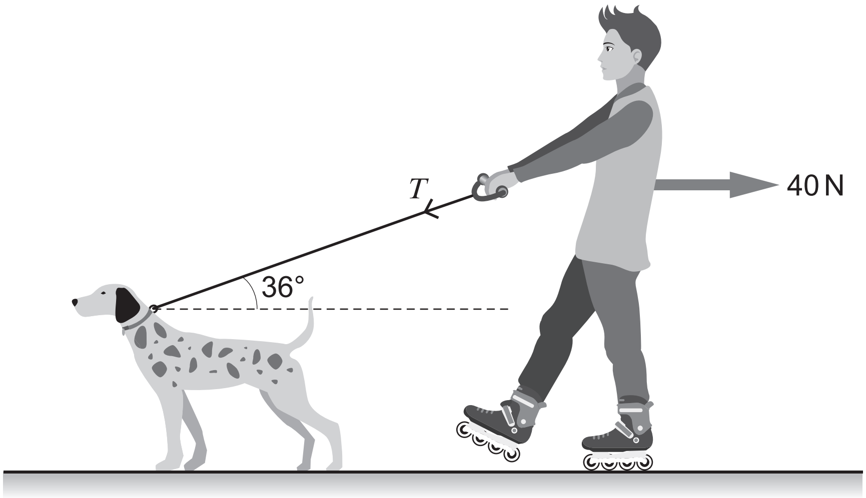
Examiner only

4. *(a)* The diagram shows a roller-skater being pulled along a horizontal road at **constant velocity**. The total resistive force to the skater's motion is 40 N.



Show that the tension, T, in the dog lead is approximately 50 N. [2]

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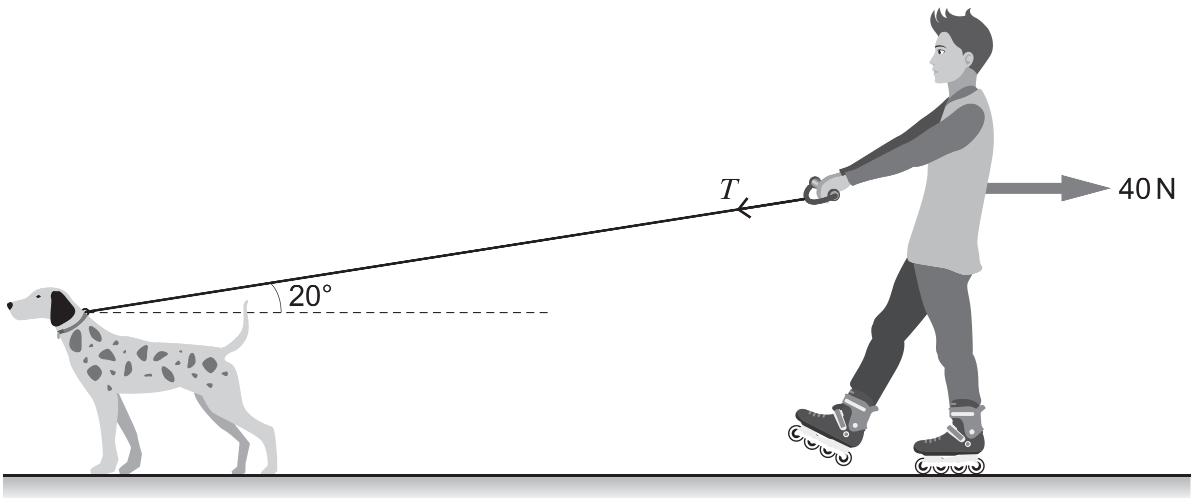
(b) Later, the roller-skater, moving at the same velocity as in part *(a)*, extends the dog lead as shown in the diagram below.



(i) Assuming that the tension, T, is the same as in part *(a)*, explain why the roller-skater initially accelerates. [1]

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Examiner only

(ii) Calculate this initial acceleration, given that the mass of the roller-skater is 35 kg. [3]

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(c) The roller-skater believes that the dog's rate of doing work is the same, regardless of the length of the dog lead. Use information from *(a)* and *(b)* to investigate whether or not this claim is correct. [3]

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